

Luca Parolari

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Summary

PhD student at the University of Padova working on self-supervised and contrastive learning in low-label and noisy settings, with a focus on medical imaging. My work involves learning robust visual representations, with a strong interest in applied computer vision and machine learning.

Personal

Birth Brescia, Italy, October 23rd 1997. Italian citizen.
Location Corteno Golgi (BS), Italy
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GitHub github.com/lparolari

Education

2023–present **PhD, Brain, Mind and Computer Science**
University of Padova, Padova, Italy

2019–2021 **Master's Degree, Computer Science and Artificial Intelligence**
University of Padova, Padova, Italy
Thesis: Weakly Supervised Visual-Textual Grounding based on Concept Similarity
Supervisor: Lamberto Ballan
Grade: 110/110 cum laude **GPA:** 28.39/30

2016–2019 **Bachelor's Degree, Computer Science**
University of Parma, Parma, Italy
Thesis: Design and Implementation in Picat of a Set Constraint Solver
Supervisor: Gianfranco Rossi
Grade: 110/110 cum laude **GPA:** 27.59/30

2011–2016 **Diploma, Computer Science**
ITCG "Olivelli-Putelli", Darfo Boario Terme, Italy
Grade: 88/100

Work Experience

01/2022–
11/2022 **AI Software Engineer**
Orobix, Bergamo, Italy
Developed AI products in the fields of computer vision and optimization planning, contributing across the full ML lifecycle from requirements gathering to deployment and maintenance. Designed full-stack applications with React, Python, and Django, and built production-ready AI infrastructures for cloud and edge environments while collaborating with stakeholders to deliver business-driven solutions.

2019–2021 **Computer Science Teacher**
"Camillo Golgi" High School, Breno, Italy
Taught computer science to upper secondary school students, developing lesson plans, programming activities, and educational materials. Prepared and evaluated written and practical exams, including serving as an examiner in the national state examination.

2019–2021 **Software Engineer**

Freelance – Esabit, Cedegolo, Italy

- **RTU & Embedded Systems.** Developed software for helicopter radio systems integrated with X-Plane, implementing GUIs and hardware interaction modules on Raspberry Pi and Arduino.
- **Asset Management Platforms.** Designed and delivered web-based asset management platforms alongside companion mobile applications for infrastructure and industrial operations.

2019–2021 **Full Stack Developer**

Freelance – Nexio Informatica, Capo di Ponte, Italy

- **Mobile App Development.** Developed React Native/Expo mobile apps for industrial operators, streamlining production line tasks, warehouse management, and work-state tracking.
- **Web Development.** Built and maintained web applications and corporate websites for external clients, managing the full software lifecycle from requirements to deployment.

2016–2019 **Full Stack Developer**

Bliss Coffee Vending, Parma, Italy

Developed web-based platforms for vending machine management, asset tracking, and financial reporting, including dashboards for operational monitoring and analytics. Built web and mobile applications for device registration, monitoring, and fiscal compliance with Italian regulations, using technologies such as PHP, Laravel, Vue.js, Java, and Android.

Academic Work Experience

2023–present **PhD Researcher in Computer Vision**

University of Padova, Padova, Italy

- **Medical imaging.** Conducted research on computer vision methods for medical imaging, focusing on analysis of endoscopic and colonoscopy data.
- **Synthetic data & evaluation.** Developed and evaluated methods for synthetic data generation and model assessment, with emphasis on robustness and error analysis of generative models.

○ Paper accepted at CVPRF 2026

○ Paper accepted at MICCAI 2025

○ Paper accepted at ISBI 2025

○ Paper accepted at ICIAP 2025

○ Paper accepted at ICPR 2024

10/2025– **Visiting PhD Student**

03/2026 *Telecom Paris – Institut Polytechnique de Paris, Paris, France*

Worked on self-supervised learning for colonoscopy video understanding. Collaborated with the LTCI Lab on research activities.

○ Paper accepted at MICCAI 2026

12/2022– **Research Fellow**

10/2023 *University of Padova, Padova, Italy*

Worked on the multi-modal models for visual grounding task using weakly-supervised learning techniques.

○ Paper accepted at BMVC 2023

2023–2025 **Teaching Assistant**

University of Padova, Padova, Italy

- **Machine Learning Course:** Delivered lectures, prepared Python/deep learning laboratory assignments, and supported project development for MSc Computer Science students.
- **Evaluation:** Assisted in grading course assignments, laboratory reports, and final examinations.

2019–2026 **Academic Service & Mentorship**

University of Padova, Padova, Italy

- **Thesis Co-Supervision:** Mentored Bachelor's and Master's students in Computer Science and Data Science on deep learning frameworks, experimental design, and scientific writing.
- **Peer Reviewing:** Serve as a technical reviewer for international journals and conferences in computer vision, machine learning, and medical image analysis.

Publications

- 2026 **L. Parolari**, P. Gori, L. Ballan, C. Biffi, L. Le Folgoc. *Contrastive Learning under Noisy Temporal Self-Supervision for Colonoscopy Videos*. Accepted at MICCAI 2026.
- 2026 **L. Parolari**, N. Faccioli, L. Ballan. *Benchmarking Layout-Guided Diffusion Models through Unified Semantic-Spatial Evaluation in Closed and Open Settings*. Accepted at CVPRF 2026.
- 2025 **L. Parolari**, A. Cherubini, L. Ballan, C. Biffi. *Temporally-Aware Supervised Contrastive Learning for Polyp Counting in Colonoscopy*. MICCAI 2025.
- 2025 E. Izzo, **L. Parolari**, D. Vezzarro, L. Ballan. *7Bench: a Comprehensive Benchmark for Layout-guided Text-to-image Models*. ICIAP 2025.
- 2025 **L. Parolari**, A. Cherubini, L. Ballan, C. Biffi. *Towards Polyp Counting in Full-Procedure Colonoscopy Videos*. ISBI 2025.
- 2024 **L. Parolari**, E. Izzo, L. Ballan. *Harlequin: Color-driven Generation of Synthetic Data for Referring Expression Comprehension*. ICPR 2024.
- 2023 D. Rigoni, **L. Parolari**, L. Serafini, A. Sperduti, L. Ballan. *Weakly-Supervised Visual-Textual Grounding with Semantic Prior Refinement*. BMVC 2023.
- 2021 **L. Parolari**. *Weakly Supervised Visual-Textual Grounding based on Concept Similarity*. M.Sc. Thesis.
- 2019 **L. Parolari**. *Design and Implementation in Picat of a Set Constraint Solver*. B.Sc. Thesis.

Computer Skills

Core	Python, PyTorch, PyTorch Lightning, Hugging Face, NumPy, Pandas, scikit-learn
Computer Vision	OpenCV, Pillow, torchvision, image/video pipelines, transformers
Infrastructure	Multi-node & multi-GPU training, SLURM, Weights & Biases, Docker, Linux
Languages	Python (primary), C/C++, Bash, TypeScript/JavaScript
Other	Git, L ^A T _E X. Full-stack background: React, React Native, Flask, Django

Languages

Italian	Native
English	Advanced / professional (C1)
French	Elementary (A2)

Interests

Ski mountaineering, cycling, hiking, astronomy.